

ALEXANDER TAMIAN

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Versatile and results-driven Senior Full Stack Engineer with 12+ years of experience delivering secure, scalable, and data-intensive applications across healthcare, education, and enterprise domains. Expert in React.js, Next.js, and Python (FastAPI, Django, Flask) with proven success designing and implementing high-performance web platforms, real-time data pipelines, and API-driven integrations.

TECHNICAL SKILLS

- **Programming Languages:**
Python, JavaScript (ES6+), TypeScript, Go, SQL, GraphQL, HTML5, CSS3, SCSS
- **Frameworks & Libraries:**
React.js, Next.js, Redux, Context API, FastAPI, Flask, Django REST Framework, Celery, Node.js, Express.js, Material UI, Tailwind CSS, D3.js, ApexCharts, Storybook, Gin, Echo
- **Databases:**
PostgreSQL, MySQL, Redis, MongoDB, DynamoDB, Cloud SQL, Firestore, Amazon Aurora, Amazon Redshift
- **Cloud Platforms & DevOps:**
AWS (Lambda, S3, RDS, ECS, EKS, API Gateway, Cognito, Bedrock), GCP (GKE, Cloud SQL, Cloud Storage, Pub/Sub, BigQuery, Firestore), Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, GitLab CI/CD
- **API Development & Integration:**
RESTful APIs, GraphQL APIs, gRPC, WebSockets, FHIR, HL7, OAuth2, OpenID Connect, JWT Authentication, API Gateway
- **Data Processing & Analytics:**
ETL Pipelines, Asynchronous Processing, Real-Time Data Streaming, Elasticsearch, Pub/Sub, Worker Threads, Node-Cron
- **Testing & Quality Assurance:**
PyTest, Jest, Cypress, Storybook, React Testing Library, Postman, Swagger/OpenAPI, SonarQube
- **Monitoring & Logging:**
Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana), Cloud Monitoring, Sentry
- **Development Practices:**
Agile (Scrum/Kanban), TDD, BDD, CI/CD, SOLID Principles, Design Patterns, CQRS, Event Sourcing, SEO Optimization, Accessibility (WCAG 2.1)

PROFESSIONAL EXPERIENCE

UNHCR, Washington

SEP 2022 – AUG 2025

Senior Full Stack Engineer

- Built a high-performance data-driven alert web app using Python with React.js, Next.js, and Express.js, implementing Amazon Cognito for secure user authentication, and utilizing Amazon API Gateway for real-time notifications, deployed on AWS EC2 with Docker.
- Designed Python-based task scheduling for processing large volumes of social media data through Meltwater API, leveraging asynchronous operations with Node-Cron and Worker Threads for optimal data throughput.
- Developed a real-time Python analytics dashboard to track and visualize social media trends, hate speech, and sentiment analysis, featuring live updates powered by Socket.IO and Python data handling techniques.

- Migrated from legacy architecture to microservices architecture on AWS, leveraging Amazon Redshift and Amazon Aurora for efficient data management and analysis, ensuring high availability and performance in a Node.js environment.
- Designed and implemented an AI-driven sentiment analysis leveraging AWS Bedrock, OpenAI's GPT-4, and advanced machine learning models to analyze real-time data, generate insights, and deliver intelligent, context-aware notifications.
- Implemented state management using Redux, Context API, and Python's data parsing capabilities to ensure efficient data handling and improved app responsiveness, supporting high-performance requirements.
- Optimized backend Python applications by applying SEO strategies, server-side rendering (SSR), and caching methods with Next.js, incorporating Python code for faster load times, lazy loading, and modular code-splitting.
- Leveraged Python for responsive data integration into web interfaces using HTML, CSS, SCSS, and Tailwind CSS, focusing on Python libraries for data manipulation and ensuring cross-browser compatibility.
- Employed Python testing frameworks like pytest alongside Jest and Storybook for unit testing and building UI components, which streamlined validation processes and enforced coding standards.
- Integrated Apexcharts with Python-generated data sets to create interactive and dynamic visualizations, enhancing user engagement and making complex data insights more accessible.
- Collaborated with data scientists to merge Python-based data models and algorithms into applications, processing and visualizing social media data from Meltwater API.
- Utilized Python with AWS services such as Lambda, API Gateway, DynamoDB, RDS, Cognito, and S3, developing scalable solutions for high-traffic, data-intensive environments.

Google, Mountain View, CA

Oct 2020 – Aug 2022

Senior Full Stack Engineer

- Developed an AI-powered advertising automation platform using React.js, Next.js, and Python (FastAPI), enabling fully automated creation, optimisation, and scaling of Google Ads campaigns for enterprise and SME clients.
- Designed and implemented autonomous ad campaign management APIs integrated with the Google Ads API, handling dynamic bidding, audience segmentation, creative testing, and keyword optimisation, reducing manual campaign management by over 40%.
- Built real-time optimisation pipelines using Google Cloud Pub/Sub, Python asyncio, and Celery to process live campaign performance data and automatically adjust bids, targeting, and budget allocation to maximise ROAS.
- Deployed scalable serverless microservices with Google Cloud Run for campaign analytics, audience segmentation, and creative generation, ensuring elastic scaling during peak ad load.
- Created scheduled automation workflows using Google Cloud Scheduler (Cron Jobs) to trigger nightly bid recalculations, budget reallocation jobs, and automated creative refreshes.
- Integrated Vertex AI with Python APIs to generate AI-driven ad copy (headlines, descriptions, keyword sets) tailored for B2B lead generation and niche industries, using fine-tuned LLM models.
- Designed data-rich analytics dashboards in React with ApexCharts and D3.js, powered by BigQuery datasets processed through Python ETL jobs, delivering actionable marketing insights like wasted spend detection and creative performance trends.
- Built a multi-channel campaign orchestration framework in Python, designed for Google Ads integration with planned expansion to LinkedIn, Meta, and TikTok APIs.
- Orchestrated ETL pipelines with Google Dataflow and Python to clean, aggregate, and transform ad performance logs into BigQuery, supporting advanced reporting and machine learning workflows.

- Secured sensitive API credentials with Google Secret Manager, stored creative assets in Cloud Storage, and implemented system health tracking via Cloud Monitoring and Cloud Logging.
- Deployed backend services on Google Kubernetes Engine (GKE) for long-running workloads and Cloud Run for serverless jobs, leveraging containerisation with Docker for portability.
- Automated CI/CD pipelines with Cloud Build and GitHub Actions, integrating pytest for backend validation, Jest/Cypress for frontend testing, and security scanning before deployments.
- Collaborated with marketing, product, and data science teams to align AI ad automation with business objectives, ensuring compliance with Google Ads policies, GDPR, and CCPA.

Newsela, New York, NY

OCT 2017 – SEP 2020

Backend Developer

- Delivered a personalized content recommendation engine using React.js, Redux, and Python (Django + Django REST Framework), leveraging user engagement data for adaptive learning experiences.
- Implemented advanced search and filtering features with Elasticsearch and PostgreSQL, enabling educators and students to quickly find relevant, leveled reading materials.
- Designed and developed dynamic lesson-building tools in React with drag-and-drop functionality, integrated with Python-based backend APIs for real-time saving and collaboration.
- Built an offline-first reading application with service workers, local storage syncing, and API reconciliation logic in Python, supporting schools with limited internet access.
- Optimized content rendering with server-side rendering (SSR) in Next.js and caching strategies to improve load times across 10M+ monthly users.
- Integrated analytics tracking with Segment and Google Analytics, processing data in Python for content performance reports and teacher dashboards.
- Migrated monolithic Django backend into modular microservices deployed on AWS ECS, with automated testing pipelines using pytest, Jest, and Cypress for end-to-end validation.
- Collaborated with editorial and curriculum teams to design scalable backend data models in Python for handling millions of articles, lesson plans, and assessments.

US Tech Solutions, Jersey City, NJ

OCT 2015 – SEP 2017

Backend Developer

- Developed enterprise HR and payroll automation tools using React.js and Python (Flask), streamlining employee onboarding, time tracking, and benefits management.
- Designed secure REST APIs in Python to integrate with external payroll systems (ADP, Paychex) and internal ERP platforms, handling sensitive PII data with encryption and role-based access control.
- Built an internal admin portal in React with reusable component libraries, real-time form validation, and data visualization dashboards powered by Chart.js and ApexCharts.
- Created scheduled data pipelines with Celery and RabbitMQ for payroll batch processing, compliance reporting, and automated tax form generation.
- Implemented authentication with JWT and Python middleware for API access control, along with audit logging for compliance with SOC 2 and GDPR.
- Deployed containerized applications with Docker to AWS EC2, setting up CI/CD pipelines with Jenkins and automated unit/integration tests with pytest and Jest.
- Improved system performance by 30% through query optimization, database indexing in PostgreSQL, and frontend state management improvements with Redux.
- Coordinated with business analysts and HR teams to translate requirements into technical specifications, ensuring alignment between engineering solutions and organizational goals.

EDUCATION

Florida State University

2014

Bachelor's degree in Computer Science